

Govt. Graduate College Muzaffargarh

Department of Information Technology



Final Year Project

Synopsis

**Title: MTNMANGOES – Online Mango Sales and
Management System**

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Abstract

The MTNMANGOES – Online Mango Sales and Management System is a web-based application designed to modernize mango trading by digitally connecting farmers and customers. Traditional mango sales in Pakistan rely heavily on middlemen, leading to reduced farmer profits, inconsistent pricing, and fragmented distribution processes. The proposed system integrates product listings, order management, secure online payments, real-time inventory tracking, and dedicated portals for farmers, customers, and administrators. By offering a transparent, scalable, and user-friendly platform, the system aims to increase farmer income, enhance customer accessibility to premium mango varieties, reduce post-harvest losses, and bring digital transformation to Pakistan's agricultural supply chain.

1. Introduction

Pakistan is among the world's leading producers of mangoes, yet the industry still follows outdated, manual, and middlemen-driven processes. These issues result in low farmer profitability and limited access for customers seeking high-quality mangoes. As digital commerce continues to expand worldwide, agricultural industries increasingly require modern systems for product distribution, online ordering, and transparent supply chain operations.

The MTNMANGOES Online Mango Sales and Management System provides a unified digital platform that enables farmers to list mango varieties, manage stock, and fulfill customer orders directly. Customers can browse fresh mangoes, place online orders, make payments, and track deliveries. Administrators manage platform operations through centralized dashboards and monitoring tools. The system increases operational efficiency, reduces manual intervention, and modernizes mango sales at a national level.

2. Problem Statement

Pakistan's mango market suffers from multiple structural and operational challenges that prevent both farmers and consumers from realizing full market potential:

- Heavy dependency on intermediaries reduces farmer profitability.
- Lack of digital selling channels limits market access for small-scale farmers.
- Inconsistent and non-transparent pricing affects buyer trust.
- Manual order handling causes delays and mismanagement.

- No centralized system exists for direct farmer–customer interaction.
- Customers struggle to access reliable, fresh produce at fair prices.
- Absence of technology adoption leads to inefficiencies and post-harvest losses.

There is a clear need for a comprehensive online platform that facilitates direct mango sales, streamlines operations, and ensures fair, transparent, and efficient transactions between farmers and buyers.

3. Objectives of the Project

Main Objectives:

- To develop a digital marketplace that enables farmers to sell mangoes directly to customers.
- To eliminate middlemen for improved farmer profitability.
- To provide secure online payment and order tracking features.
- To ensure real-time inventory management and product updates.
- To offer dedicated portals for customers, farmers, and administrators.
- To centralize system data for transparency, reporting, and analytics.

4. Scope of the Project

Included in Scope:

- Customer registration, login, and product browsing.
- Online mango ordering and secure payment integration.
- Order tracking and customer feedback.
- Farmer account creation, product listing, and inventory management.
- Admin module for user, order, and product oversight.
- Reporting and analytics dashboard.
- Role-based access control.

Excluded from Scope:

- Mobile application development.
- Integration of external logistics companies.
- Automated quality grading using AI.
- Export licensing or international shipment management.

5. Literature Review

A review of existing research and systems reveals both the potential and the current gaps in digital agricultural commerce, particularly for fruit marketing in developing economies.

1. Ali et al. (2020) – Digital Agriculture Marketplaces in South Asia

Ali et al. examined the adoption of e-commerce platforms among small-scale farmers in South Asia, finding that direct-to-consumer digital channels increased average farmer income by 23% while reducing post-harvest losses by 18%. Their study identified the absence of specialized crop-specific marketplaces as a key barrier to wider adoption. MTNMANGOES directly addresses this gap by providing a mango-specific platform tailored to the Pakistani agricultural context.

2. Khan & Raza (2019) – Middlemen and Price Distortion in Pakistan's Mango Supply Chain

This study analyzed the economic impact of commission agents (arhtis) in Punjab's mango markets. The authors found that intermediary commissions consumed 30–40% of retail value before income reached growers, and that price opacity led to significant consumer overpayment. The study recommended technology-driven disintermediation as the most viable solution, a recommendation the proposed system directly implements.

3. Ahmad et al. (2021) – E-Commerce Adoption in Pakistani Agriculture

Ahmad et al. surveyed 350 farmers across Multan, Bahawalpur, and Rahim Yar Khan regarding their willingness to adopt online selling platforms. Results showed 72% expressed interest in direct online sales but cited lack of platform availability and digital literacy as primary obstacles. The study validates the market need for MTNMANGOES and underscores the importance of a simple, user-friendly interface.

4. Zhao & Li (2018) – Agricultural E-Commerce Platforms: A Global Review

This comparative study reviewed 12 fruit-trading platforms from China, India, Morocco, and Brazil. The authors concluded that platforms combining farmer listing management, consumer ordering, real-time inventory tracking, and integrated payment gateways achieved 3× higher transaction volumes than partial-feature alternatives. The MTNMANGOES system architecture mirrors this successful multi-module approach.

5. Iqbal et al. (2022) – Post-Harvest Losses in Pakistan's Fruit Sector

Iqbal et al. estimated that Pakistan loses approximately 35–40% of its annual mango harvest due to inefficient supply chain management, inadequate storage, and delayed distribution. The study highlighted that digitizing order fulfillment and enabling direct farmer–buyer communication could cut losses by up to 25%. MTNMANGOES addresses these recommendations through its real-time inventory and order management modules.

Collectively, this body of literature confirms both the market need and technical viability of MTNMANGOES. Existing platforms lack the crop-specific focus, integrated payment systems, and multi-stakeholder architecture that the proposed system provides.

6. Proposed System / Methodology

System Overview:

The MTNMANGOES system integrates customer, farmer, and admin modules into one unified platform that manages listings, transactions, inventory, and reports. All modules share a common database layer, ensuring data consistency and real-time updates across user roles.

Methodology / Approach:

SDLC Model: Waterfall Model

Technologies:

- Frontend: HTML5, CSS3, JavaScript
- Backend: PHP (Laravel Framework)
- Database: MySQL
- Payment Gateways: Stripe, PayPal
- Tools: XAMPP, VS Code, phpMyAdmin, GitHub, Postman

Functional Requirements:

- Customer registration, login, ordering, and payment.
- Farmer product listing and stock management.
- Order processing workflow for farmers and admins.
- Admin dashboard for platform control.
- Inventory and order tracking.
- Report generation.

Non-Functional Requirements:

- Performance
- Security
- Scalability
- Usability
- Reliability

7. Feasibility Analysis

Technical Feasibility:

The project uses widely supported, open-source technologies such as Laravel, MySQL, and JavaScript, ensuring smooth implementation without proprietary software dependencies.

Economic Feasibility:

The system is cost-effective due to minimal development expenses and zero licensing fees. Hosting and domain costs remain affordable on shared or cloud infrastructure.

Operational Feasibility:

The system simplifies operations for farmers, customers, and administrators. It significantly reduces manual workload and ensures fast, accurate transactions through automated workflows.

8. Expected Outcomes

- A fully functional online mango selling and management system.
- Increased farmer profit through elimination of middlemen.
- Digital marketplace with customer ordering and payment features.
- Automated inventory and order management.
- Enhanced transparency in mango trading.
- Centralized reporting and analytics dashboard.
- Improved customer accessibility to fresh, premium mangoes.

9. Work Plan / Timeline

The following table outlines the major phases, their durations, and milestone weeks:

Phase	Original Duration	Expanded Duration	Weeks
Requirement Gathering	1 week	3 weeks	1 – 3
System Design	2 weeks	5 weeks	4 – 8
Development	5 weeks	14 weeks	9 – 22
Testing & Debugging	2 weeks	6 weeks	23 – 28
Final Documentation & Submission	2 weeks	6 weeks	29 – 34